

Math 10A with Professor Stankova

Quiz 11; Wednesday, 11/8/2017

Section #106; Time: 10 AM

GSI name: Roy Zhao

Name: _____

Circle True or False or leave blank. (1 point for correct answer, -1 for incorrect answer, 0 if left blank)

1. True **FALSE** The median of a continuous distribution may not exist.

Solution: The median always exists but the mean may not exist.

2. True **FALSE** In order to determine the median of the distribution $f(x) = e^{-x^2}/\sqrt{\pi}$, we must calculate the CDF and set it equal to $1/2$.

Solution: Since the distribution is symmetric, we automatically know that the median is 0.

Show your work and justify your answers. Please circle or box your final answer.

3. (10 points) (a) (2 points) Suppose that $f(x) = \begin{cases} -Cx^{-3} & x \leq -1 \\ 0 & x > -1 \end{cases}$ for some constant C . If f is a PDF, describe how you would find C (you don't have to solve).

Solution: Since f is a PDF, we require that

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{-1} -Cx^{-3}dx = \left. \frac{-Cx^{-2}}{-2} \right|_{-\infty}^{-1} = \frac{C}{2} = 1.$$

Therefore $C = 2$.

- (b) (2 points) If you solve, you get $C = 2$. Find the CDF of f from above.

Solution: We split up our CDF as

$$F(x) = \begin{cases} \int_{-\infty}^x -2/t^3 dt & x \leq -1 \\ \int_{-\infty}^{-1} -2/t^3 dt & x \geq -1 \end{cases} = \begin{cases} x^{-2} & x \leq -1 \\ 1 & x \geq -1 \end{cases}.$$

- (c) (3 points) Find the median of this distribution.

Solution: The median is when $F(x) = 1/2$ or when $1/x^2 = \frac{1}{2}$ and $x = \pm\sqrt{2}$. We choose $x = -\sqrt{2}$ in order to get the correct negative interval.

- (d) (3 points) Find the mean of this distribution.

Solution: We take the integral

$$\mu = \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{-1} -2/x^{-2}dx = 2/x \Big|_{-\infty}^{-1} = -2.$$